

Team 6: Product Requirements Document

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Problem Statement

Consumers are increasingly looking for healthier alternatives to ultra-processed foods, and demand for organic groceries has skyrocketed. According to the Organic Trade Association, Americans organic sales reached a record \$71.6B in 2024 and has been achieving record growth year after year.

Downtown San Jose faces barriers that make organic groceries hard to access. The area's high population density, traffic congestion, and limited parking make driving to specialty grocery stores inconvenient, especially during peak hours. This issue is amplified by the large student population at San José State University (SJSU) in the heart of downtown, where many students don't have cars and instead rely on public transit or walking. This limits how often they can visit these stores as public transit is slow and their schedules are busy and limits their purchasing power as they have to carry their purchases manually back home.

OFS is a new organic retailer in Downtown San Jose. OFS's mission is to improve access to quality foods by providing a reliable online ordering and home delivery service for organic groceries. OFS plans to use a fleet of self-driving robotic delivery vehicles to reduce the time spent shopping in a dense, traffic-congested downtown area. Users simply have to pick up their orders and the robots drop off at their front porch. However, OFS currently lacks the digital infrastructure needed to support this service, including a website where customers can browse and purchase items and a backend system to manage the robotic delivery fleet.

Proposed Solution

The customer-facing website must be reliable and kept up to date with OFS's current inventory. It must securely store customer information, including account details and delivery addresses, and protect that data using appropriate authentication and security measures. The interface should be intuitive and easy to navigate, providing a search bar for finding specific items and clear categories such as fruits, vegetables, and other grocery items. The user interface should allow customers to purchase their goods in the shortest time possible. Features that support the ability to schedule automatic weekly orders and a "recently ordered" section.

When placing an order, customers can choose to pay an extra fee for home delivery. This fee is determined by the total order weight. If the order weighs less than 20 pounds the delivery is free, otherwise a \$10 delivery charge is added at checkout. To support this, the system must calculate total cart weight using item weights stored in the database and add the final cost

with the order cost. If there are multiple items the total weight is the sum of the weights of all items multiplied by that item's weight. After a purchase is made an update has to be applied to the database to recalculate the total inventory and OFS's revenue.

On the operations side, the system must schedule and coordinate OFS's delivery robots. Each robot trip can carry up to 10 different orders and up to 200 pounds total, so the system must group incoming orders into trips without exceeding these limits. For each trip, the system must generate an optimized delivery route using Google Maps, ordering stops efficiently and accounting for traffic when possible. These deliveries must be strategically planned by the system's routing algorithm to reduce OFS operating costs and ensure groceries are delivered safely, efficiently, and on time.

Additionally, the system must include a separate employee and manager portal where OFS staff can log in and access a dashboard to query and manage the inventory database. Changes made in the inventory database should be reflected in the customer-facing website.

Detailed Breakdown:

Personas:

- **Lucas, SJSU Student:** Lucas is a 21 year-old nutrition major at San Jose State University who lives in the dorms and shops on a moderate student budget. He is health-focused and wants affordable organic food, but he does not have a car and typically relies on a bike for transportation. This makes trips to specialty groceries inconvenient and limits how many groceries he can carry. Between his studies and spending time with friends, Lucas rarely has time for long shopping trips and only gets quick healthy snacks throughout the day. A reliable OFS delivery service would allow him to consistently access organic groceries without sacrificing time or dealing with carrying groceries on his cheap bike. For Lucas, nothing beats coming back from a long day of classes and finding a healthy snack at home that he's been craving all day.
- **Sophia, Busy Tech Professional:** Sophia is a 25 year-old software engineer living in a penthouse in one of Downtown San Jose's skyscrapers. She works demanding hours and is always on call. She earns \$500,000 per year at Adobe and is willing to splurge on premium services. Sophia only gets a small amount of time off a week and wants to dedicate that time to doing things that she loves, not waiting in line at a supermarket. OFS's robotic delivery system might empower Sophia to efficiently spend time on things that matter most to her. For Sophia, OFS's system should be as effortless and seamless as possible, with cost being a secondary concern.
- **John, Family Parent Shopper:** John is a 38 year old father who works as an accountant at a local firm. Being married with two children, John is very budget-conscious and prioritizes financial stability both at his workplace and at home. He and his wife have

the household expenses which includes groceries. They prefer shopping weekly for fresh items in bulk in order to maximize their savings and reduce the amount of time spent on errands. With work and family commitments, both John and his wife have tight and organized schedules for certain activities. Their purchasing decisions have cost and convenience factors, ensuring having stocked food without exceeding family budget. With the development of OFS, John has an alternative for grocery shopping and household management. OFS offers an environment where John can place grocery orders without him having to spend time at a store. The OFS delivery service robot allows for quick and direct delivery at his doorstep at a convenient time. This allows him to manage other responsibilities, while also ensuring his family's house pantry is fully stocked.

- **Robert, Senior Resident:** Robert is a 85 year-old retired senior who lives alone in Downtown San Jose and has limited mobility. Robert played football in college but had to quit due to a leg injury. He is too senile to drive and has difficulty walking long distances and carrying shopping bags, which makes grocery shopping in person a challenging and sometimes unsafe. Robert struggles to navigate complex websites and prefers sites that are minimalistic and that have clear well defined buttons. OFS's delivery system can make something that was once a monumental challenge for Robert much easier that doesn't require him to leave his porch.

Scenarios:

- **Lucas, College Student Quick Order Scenario:**
Lucas is a college student who goes to SJSU as a nutritionist major and lives on campus for the entire year because he is not from the Bay. He notes that the school's Dining Commons building and other food courts are closed during the weekends, making him more conscious of the food he has in his dorm. Living on campus means he doesn't have access to a vehicle, and walking to a grocery store is not very convenient. Rather than relying on expensive takeout from DoorDash or UberEats, Lucas has another alternative: OFS, a grocery delivery robot service.

He logs into his OFS account from his laptop in his dorm room and navigates through the "Fresh Produce" category. Since he is on a tight student budget and only needs lightweight items he can easily store in his mini-fridge, he uses the app's filters to filter by price and selects the "Under \$5", "Organic", and "Low Weight" options. He selects a box of cereal, bananas, and granola bars as they are simple yet healthy options that will last through the weekend, and adds them to his virtual shopping cart.

Because he needs the groceries as soon as possible, he chooses the "same day delivery" option and pays securely online using his debit card. Shortly after completing his purchase, he receives a confirmation email with live-tracking information and delivery details. Later that day, the OFS robot arrives outside his dorm building and successfully

